Article

Explainable El Niño predictability from climate mode interactions

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The El Niño–Southern Oscillation (ENSO) provides most of the global seasonal climate forecast skill[1–3], yet, quantifying the sources of skilful predictions is a long-standing challenge[4–7]. Different sources of predictability affect ENSO evolution, leading to distinct global effects. Artificial intelligence forecasts offer promising advancements but linking their skill to specific physical processes is not yet possible[8–10], limiting our understanding of the dynamics underpinning the advancements. Here we show that an extended nonlinear recharge oscillator (XRO) model shows skilful ENSO forecasts at lead times up to 16–18 months, better than global climate models and comparable to the most skilful artificial intelligence forecasts. The XRO parsimoniously incorporates the core ENSO dynamics and ENSO's seasonally modulated interactions with other modes of variability in the global oceans. The intrinsic enhancement of ENSO's long-range forecast skill is traceable to the initial conditions of other climate modes by means of their memory and interactions with ENSO and is quantifiable in terms of these modes' contributions to ENSO amplitude. Reforecasts using the XRO trained on climate model output show that reduced biases in both model ENSO dynamics and in climate mode interactions can lead to more skilful ENSO forecasts. The XRO framework's holistic treatment of ENSO's global multi-timescale interactions highlights promising targets for improving ENSO simulations and forecasts.

The El Niño–Southern Oscillation (ENSO) exerts global environmental and socioeconomic effects by means of teleconnections[1–3]. Since the first successful prediction of El Niño in 1986 (ref. 4), decades of progress on the understanding and modelling of ENSO have improved prediction skill[5–7]. However, skilful prediction of ENSO at a lead time longer than 1 year remains a challenge.

Whereas ENSO originates from coupled ocean–atmosphere interactions in the tropical Pacific, recent studies highlight that interactions with other ocean basins could potentially improve ENSO prediction[11]. For instance, many other climate modes have been shown to interact with ENSO (Fig. 1a), including the North and South Pacific Meridional Modes (NPMM and SPMM)[12,13]; the Indian Ocean Basin (IOB) mode[14], the Indian Ocean Dipole (IOD) mode[15] and the Southern Indian Ocean Dipole (SIOD) mode[16] in the Indian Ocean (IO); as well as Tropical North Atlantic (TNA) variability[17], the Atlantic Niño (ATL3)[18] and the South Atlantic Subtropical Dipole (SASD) mode[19] in the Atlantic Ocean (AO). Although several previous studies designed forecast experiments to illustrate the roles of other ocean basins in ENSO predictability, using simple coupled models[14,20,21], atmosphere–ocean coupled general circulation models (CGCMs)[22–26] or linear inverse models[27,28], it remains a challenge quantifying the relative contributions of other ocean basins to ENSO predictability. The CGCMs used typically show pronounced biases in simulating both the climate mean state and ENSO dynamics, thus hindering skill in predicting ENSO and complicating quantification of the other ocean basins' effect on ENSO predictability. Current linear inverse models are by construction not able to fully capture ENSO's nonlinear dynamics and seasonality[27,28]. Quantifying the sources of skilful predictions from these specific physical processes has been elusive[11,15,17,29,30].

Different sources of ENSO predictability can lead to substantial event-to-event differences in ENSO evolution and associated global impacts. For example, although both the 1997–1998 and 2015–2016 extreme El Niño events had similar amplitudes of Niño3.4 sea-surface temperature anomalies (SSTAs), they had distinct precursor patterns (Fig. 1b). The 1997–1998 event showed strong preconditioning by means of recharged warm water volume (WWV) in the equatorial Pacific, large SSTA precursors in the IO (including a negative IOD during 1996 September–October–November), but only weak SSTAs in the extratropical Pacific. By contrast, the 2015–2016 event was characterized by a weaker build-up of WWV, less-pronounced precursor SSTAs in the IO

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